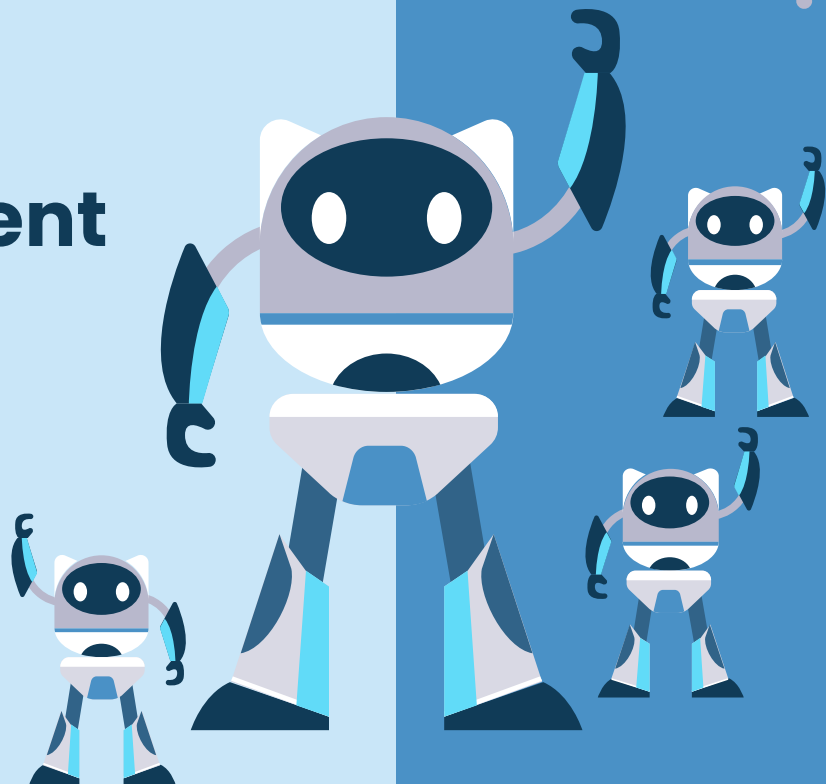


Conflict-Based Search for Optimal Multi-Agent Pathfinding

Presented by Andrea Turek
(Sharon et al., 2015)



The Problem: MAPF



Multiple robots



Shared environment



Collision-free paths



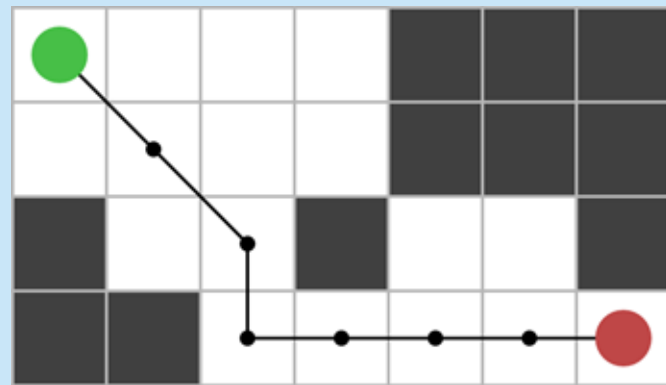
Minimize total costs



A* in MAPF Problems

Why not just run joint A*?

- For one robot A* works well
- For two robots, state is their joint position
- Every move considers all combinations of both robots actions
- Branching factor grows exponentially in number of robots



2 robots $\rightarrow$ 25 combinations

3 robots $\rightarrow$ 125 combinations

k robots $\rightarrow 5^k$ combinations

Coupled vs. Decoupled Methods

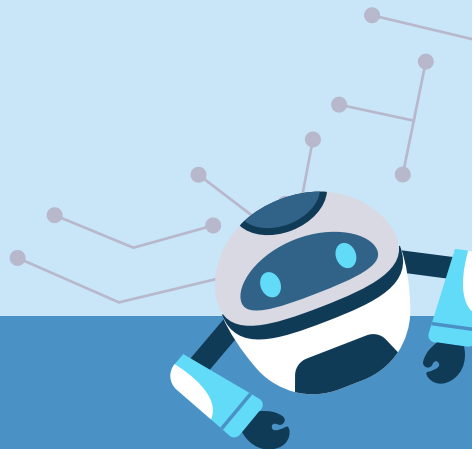
Coupled

- Search joint state space
- Complete & optimal
- Very expensive
- Giant state space with exponential complexity

Decoupled

- Plan all robots separately
- Faster
- Scales better
- May fail or be suboptimal

Can we couple robots only when necessary?



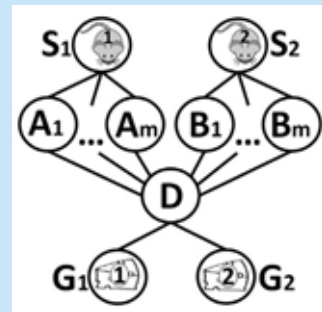
Main idea of CBS

General idea

- Conflict-based search is a **complete** and **optimal** MAPF solver
- Plan for each robot individually & only couple when needed
- Functions at two levels (high and low)

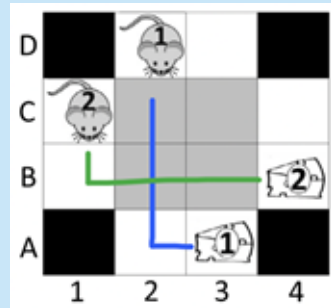
1) High Level

- Constraint tree
 - **Node** = set of constraints
 - **Edge** = resolve a conflict
 - **Leaf** = conflict free solution
- Reasons about multi-agent conflicts

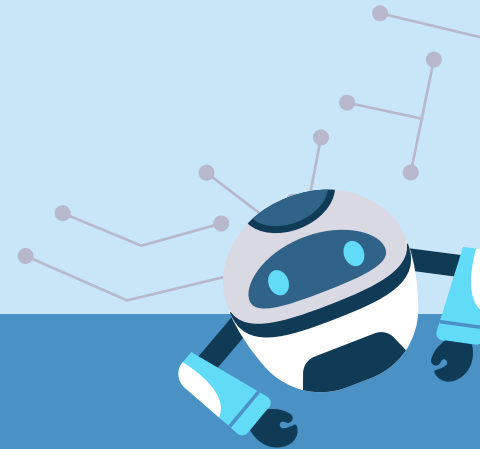
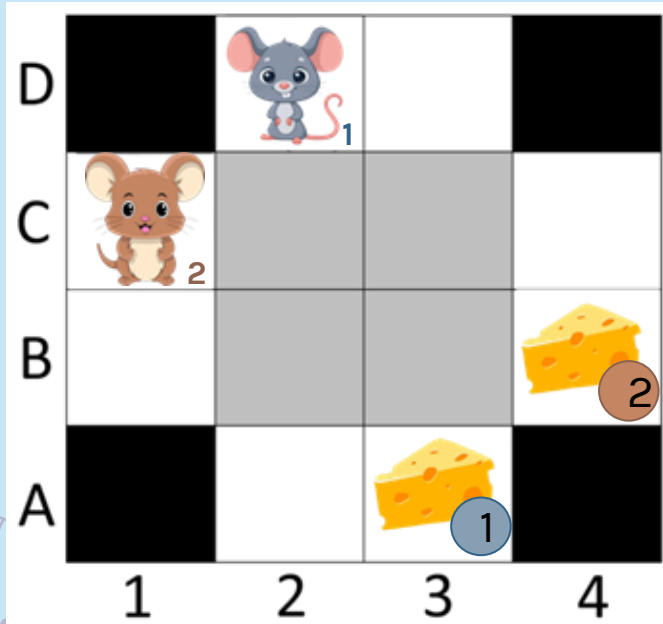


2) Low Level

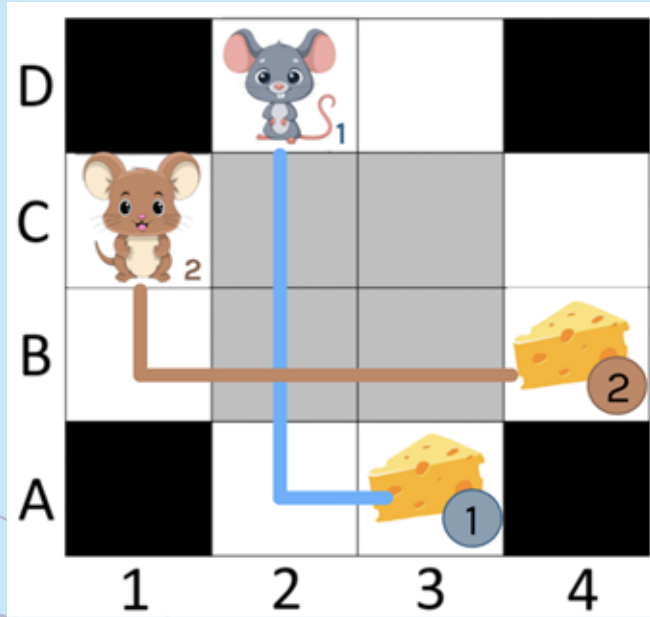
- Input: constraints for that robot
- Finds shortest path satisfying constraints
- Only replans the robot involved in the conflict



CBS Problem



CBS Problem



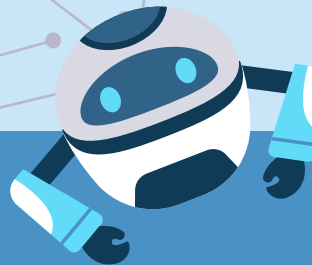
Con: 0

Cost: 8

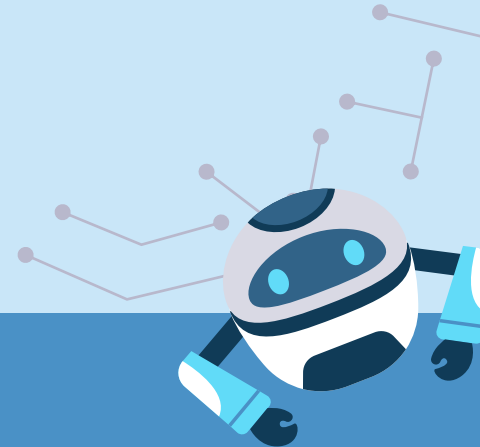
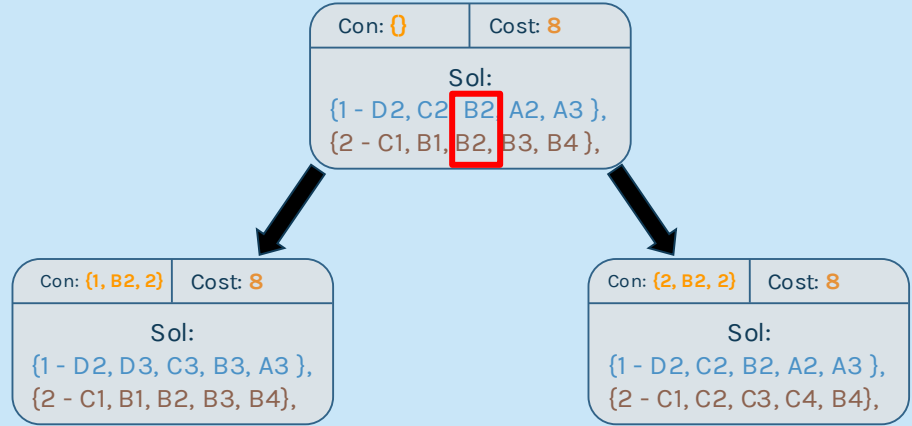
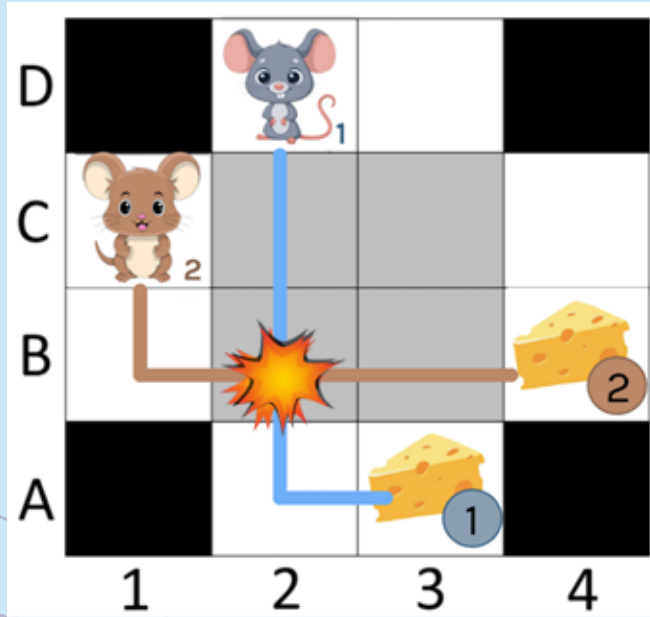
Sol:

{1 - D2, C2, B2, A2, A3 },

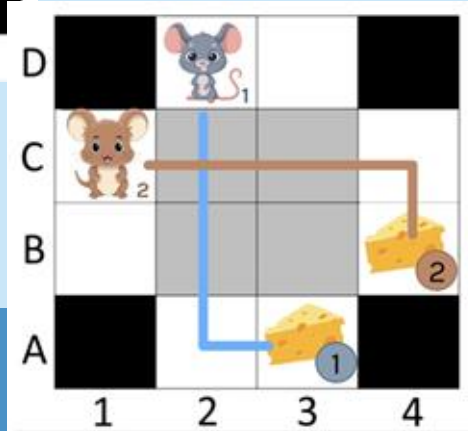
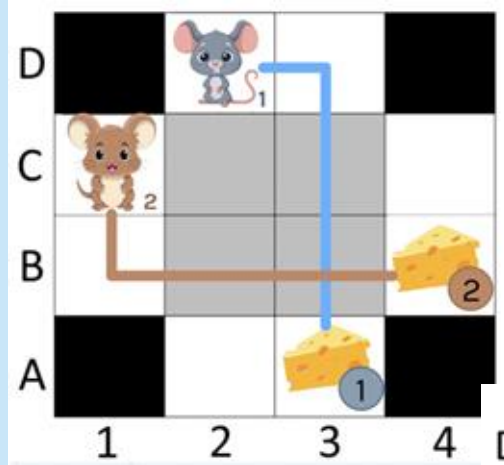
{2 - C1, B1, B2, B3, B4},



CBS Problem



CBS Problem



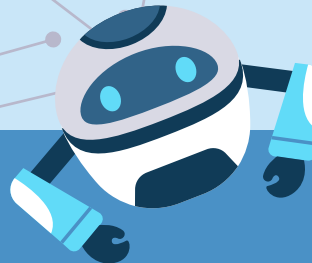
Mouse 1 cannot be at B2 @ $t = 3$

Con: 0	Cost: 8
Sol: {1 - D2, C2, B2 , A2, A3 }, {2 - C1, B1, B2, B3, B4 },	

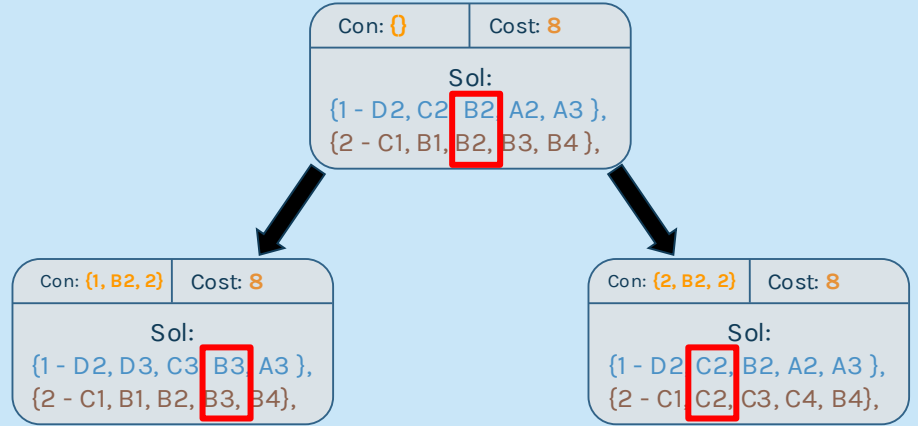
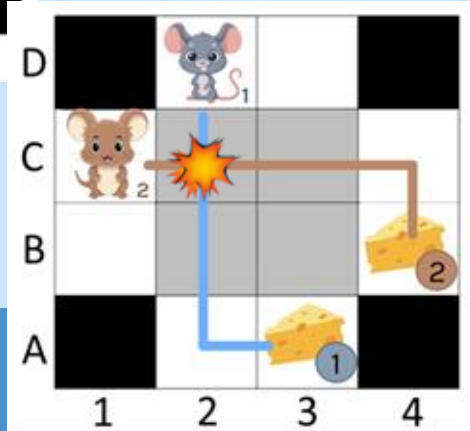
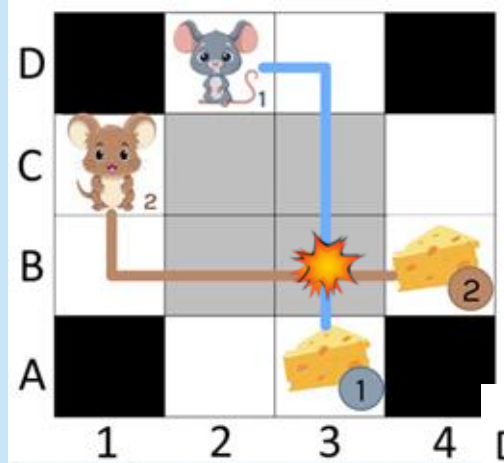
Con: {1, B2, 2}	Cost: 8
Sol: {1 - D2, D3, C3, B3, A3 }, {2 - C1, B1, B2, B3, B4 },	

Con: {2, B2, 2}	Cost: 8
Sol: {1 - D2, C2, B2, A2, A3 }, {2 - C1, C2, C3, C4, B4 },	

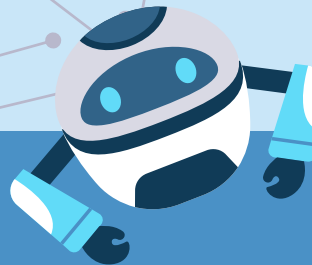
Mouse 2 cannot be at B2 @ $t = 3$



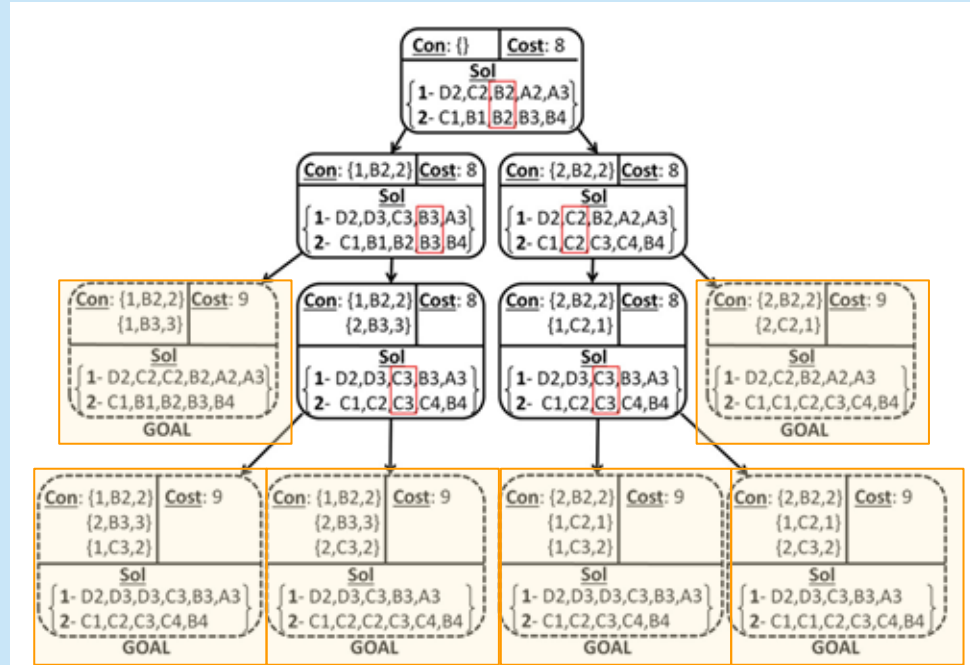
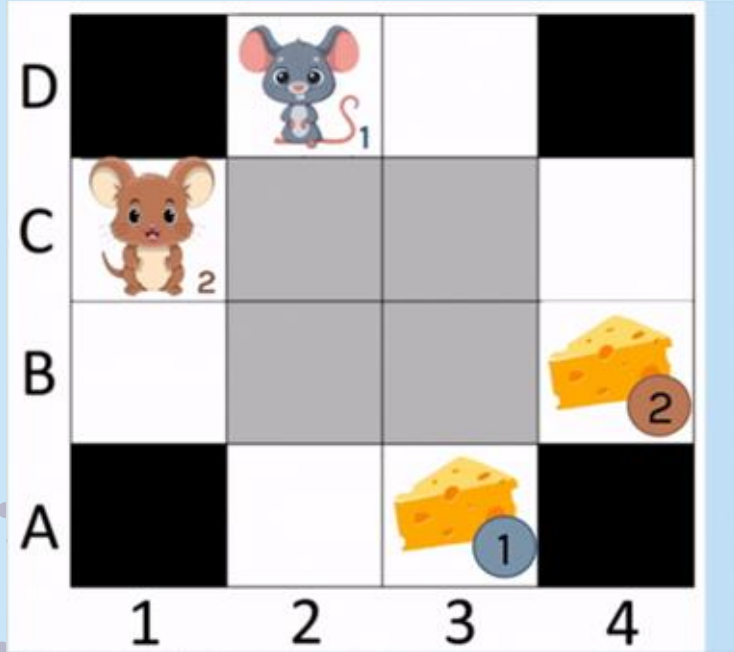
CBS Problem



Continue branching until you find a
collision free solution

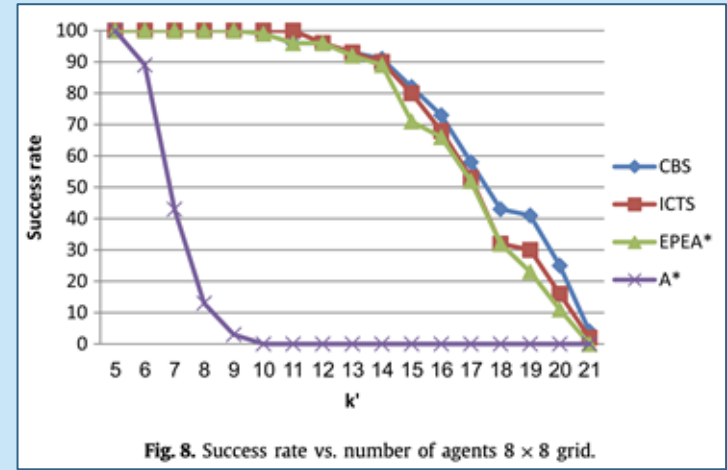


CBS Problem

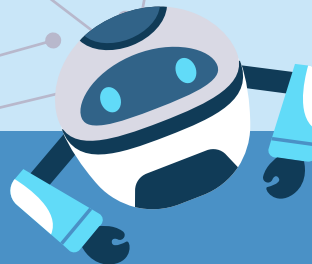


Why CBS Works

- Complete and optimal
 - High-level search finds best first total cost
 - Low-level finds optimal single agent-path
- Only branches when a conflict appears
 - Avoids exploring full joint space
- If robots don't interact, planned independently
- If they do interact, add one constraint to resolve it

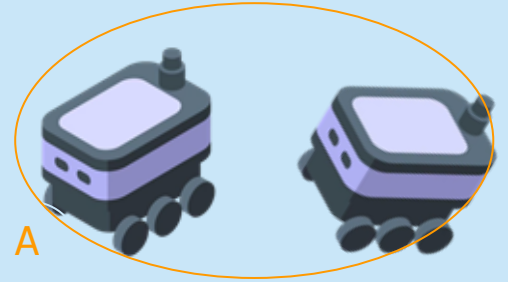


A* degrades rapidly, CBS maintains high success with many robots



Extensions (Meta-Agent CBS)

- Sometimes two robots conflict over and over
 - Parameter B controls merging
 - If collisions $> B$ merge
- Repeated conflicts $\rightarrow$ merge agents
- Solve merged group jointly



B

Robots A & B collide many times in our environment

$c > B \rightarrow$ We merge them



Results

CBS complexity depends on the density of conflicts, not directly on the number of robots.

Structured Local Bottlenecks

- Conflicts are **localized**
- Conflicts are more predictable
- Only a few robots interact at a time
- Constraint tree stays shallow , **CBS performs well**



Entangled Open Interactions

- Many robots interact **simultaneously**
- Resolving a conflict creates others
- Constraint tree grows rapidly, **CBS struggles**

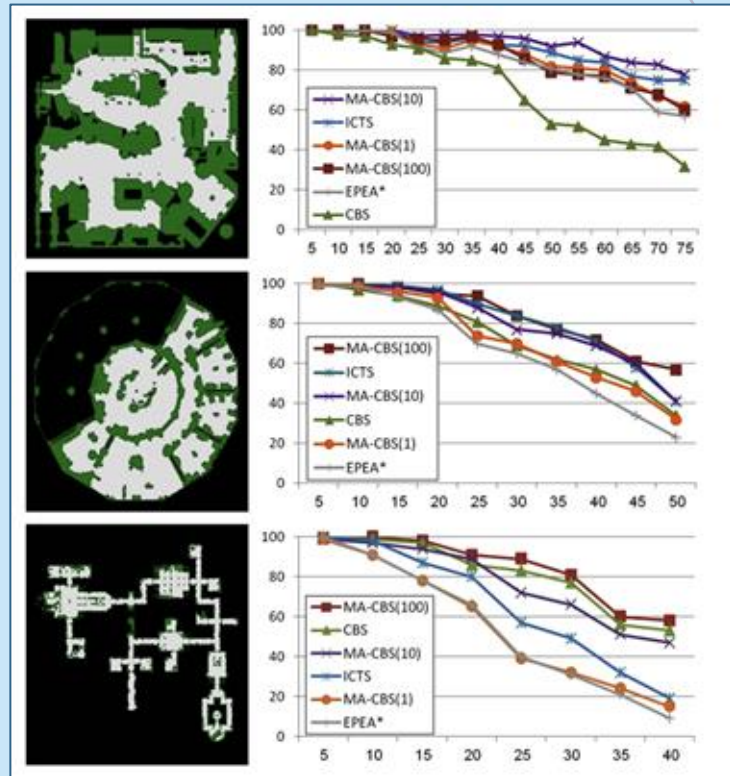


Big Picture Takeaway

- CBS works well when conflicts are localized
- MA-CBS improves performance when conflicts are frequent & repeated
- The right level of merging depends on conflict density & map structure

CBS-related works in our lab

- CBS-MP
 - Extends CBS to motion planning in continuous space
- TMP-CBS
 - CBS used to coordinate both motion and task decisions
- Hierarchical Composition CBS (HCCBS)
 - Scales CBS by solving smaller agent groups first and hierarchically merging them, reducing the size of the conflict tree for large, crowded problems



Success Rate of Algorithms Vs. Robot Count in different Environments

Thank you

Questions?

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